

Ecosystem Transpiration and Evaporation: Insights from Three Water Flux Partitioning Methods across the OzFlux Network

(A Replication of Nelson et al., 2020 for Australian Ecosystems)

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Abstract

This report replicates the core analysis framework of Nelson et al. (2020) *Global Change Biology*, applying it to 45 sites across the Australian OzFlux network. We evaluate three ecosystem evapotranspiration (ET) partitioning methods: the Transpiration Estimation Algorithm (TEA), the Zhou/uWUE method, and the Pérez-Priego method. While the three methods show strong temporal correlation, we observe significant divergence in the absolute magnitude of the transpiration-to-evapotranspiration fraction (T/ET). TEA consistently estimates higher T/ET (network mean ~ 0.60), while Zhou and Pérez-Priego yield more conservative estimates (~ 0.40). We investigate these distributions across IGBP biomes, temporal dynamics, and primary environmental drivers including Leaf Area Index (LAI: 2.07 ± 0.35 - $6.10 \text{ m}^2/\text{m}^2$) and Aridity Index (AI: 0.64 ± 0.11 - 2.69).

1 Introduction

Partitioning total ecosystem evapotranspiration (ET) into transpiration (T) and soil/canopy evaporation (E) is critical for understanding water-carbon coupling. Nelson et al. (2020) demonstrated that while partitioning algorithms broadly agree on the temporal dynamics of transpiration, their differing assumptions lead to significant divergence in absolute T/ET magnitudes. In this analysis, we apply the exact methodology to the OzFlux network spanning diverse Australian ecosystems.

2 Methodology

2.1 Data

Eddy covariance flux data (Level 6) from 45 OzFlux sites were utilized, spanning diverse Köppen climate zones. We extracted variables including ET, GPP, VPD, R_g , and Tair. Leaf Area Index (LAI) was sourced from MODIS MOD15A2H climatology (mean: $2.07 \text{ m}^2/\text{m}^2$), and Aridity Index (AI = P/PET) was directly computed from site-level meteorological drivers (mean: 0.64).

2.2 Partitioning Methods

1. **TEA (Nelson et al. 2018)**: A machine learning approach (Random Forest) trained exclusively on dry-canopy periods where E is assumed negligible.
2. **Zhou / uWUE (Zhou et al. 2016)**: Derived from the underlying water-use efficiency theory, relating GPP and VPD to stomatal conductance.
3. **Pérez-Priego (2018)**: An alternative physiological threshold-based approach.

3 Results

3.1 Spatial and Biome Distribution

The 45 OzFlux sites span a wide gradient of aridity (AI: 0.11-2.69) and vegetation structures (LAI: 0.35-6.10 m²/m²). Figure 1 shows their spatial distribution.

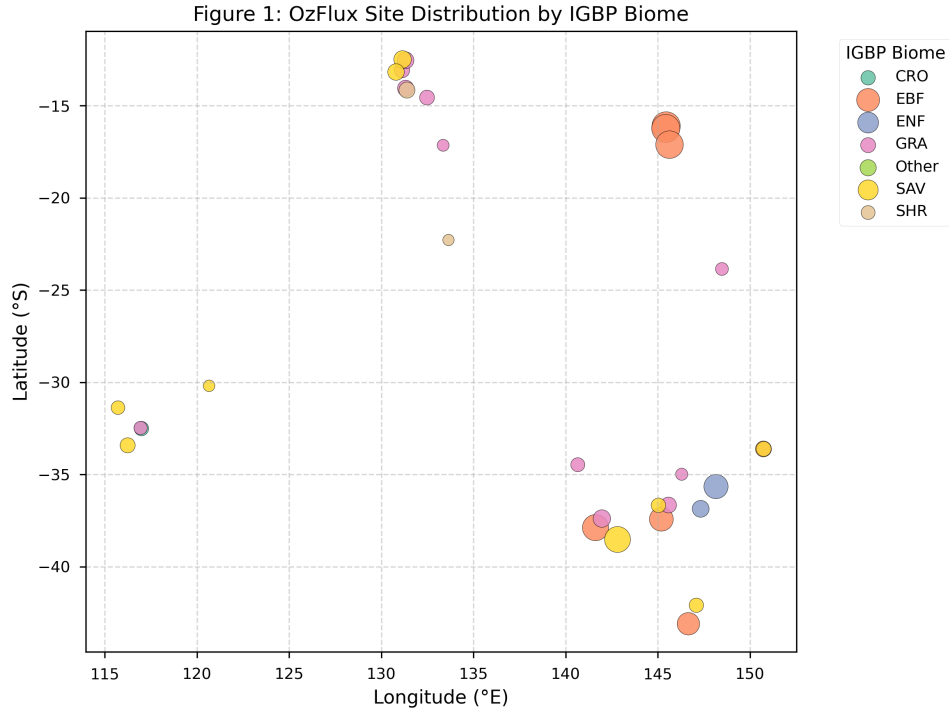


Figure 1: Distribution of the 45 OzFlux sites colored by their primary IGBP biome classification and sized by LAI.

3.2 T/ET Distribution by Biome

Replicating Figure 2 from Nelson et al. (2020), we find that TEA consistently estimates the highest T/ET ratios across almost all biomes, particularly in forested regions (EBF, ENF). Savannas and Grasslands exhibit the widest inter-method spread, reflecting the role of bare soil evaporation in lower LAI ecosystems.

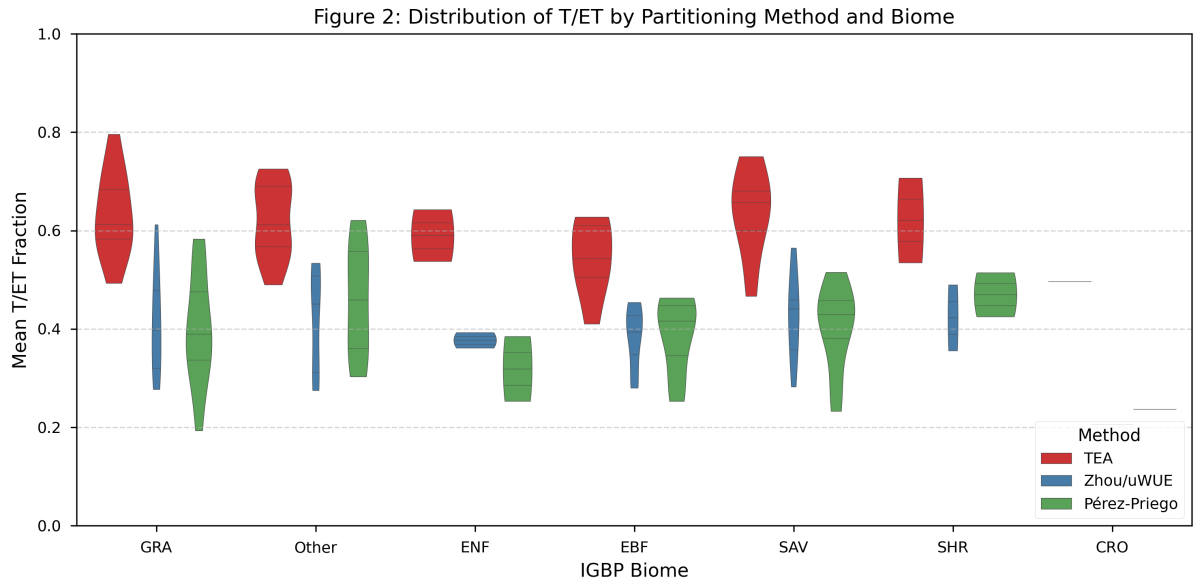


Figure 2: Probability density (violin plots) of T/ET for TEA, Zhou, and Pérez-Priego across IGBP biomes.

3.3 Inter-method Agreement

Figure 3 replicates the daily scatter analysis. The methods exhibit moderate to high temporal correlations ($r = 0.50$ to 0.80), indicating they respond similarly to short-term environmental drivers (VPD, radiation). However, the Zhou and Pérez-Priego methods are systematically biased lower compared to TEA (shown by the deviation from the 1:1 line).

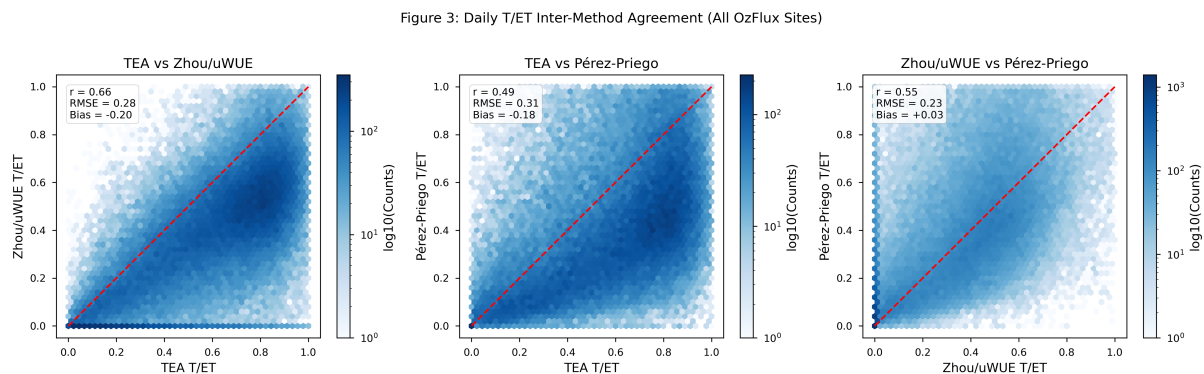


Figure 3: 2D Hexbin density plots comparing daily T/ET estimates across the three methods. The red dashed line represents the 1:1 relationship.

3.4 Seasonal Dynamics

Figure 4 illustrates the mean monthly seasonal cycle of T/ET for the four most populated biomes in the dataset.

Figure 4: Seasonal Climatology of T/ET by Biome

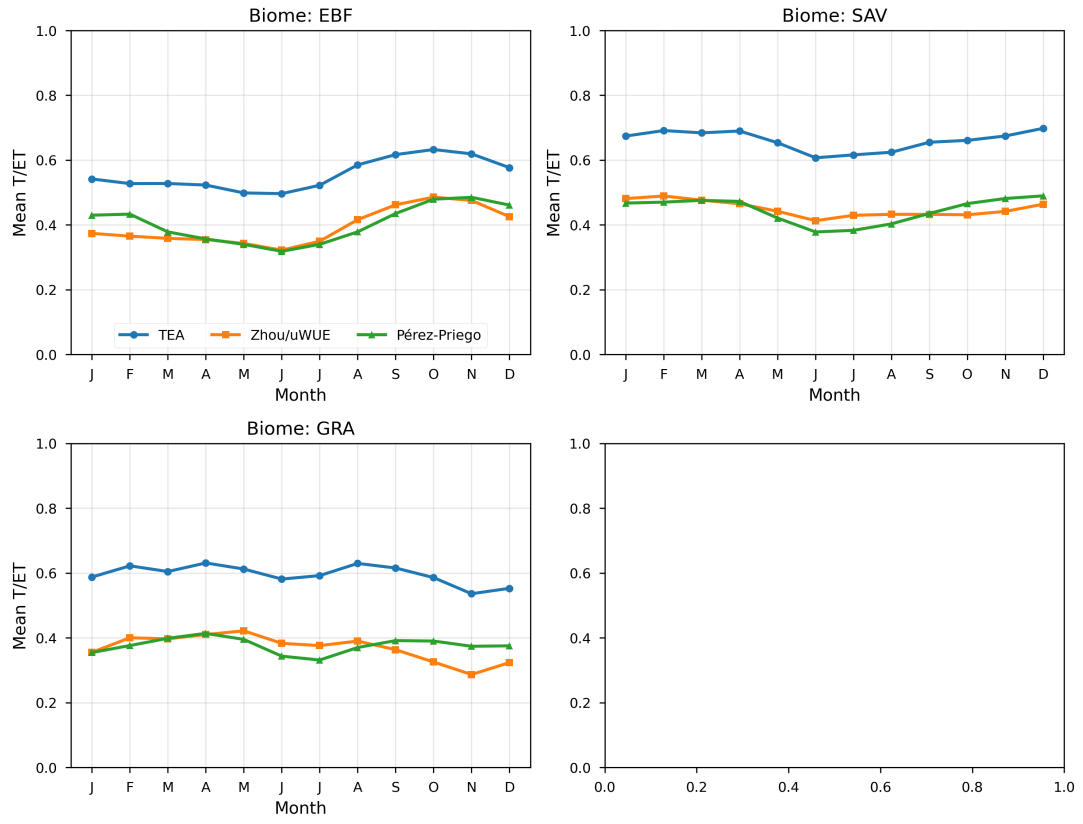


Figure 4: Mean monthly seasonal cycle of T/ET for the four most populated biomes in the dataset.

3.5 Environmental Drivers

Following Nelson et al., Figure 5 isolates the primary environmental drivers of the mean site-level T/ET ratio: Aridity Index (AI) and Leaf Area Index (LAI).

Figure 5: Environmental Drivers of T/ET (Aridity Index vs Leaf Area Index)

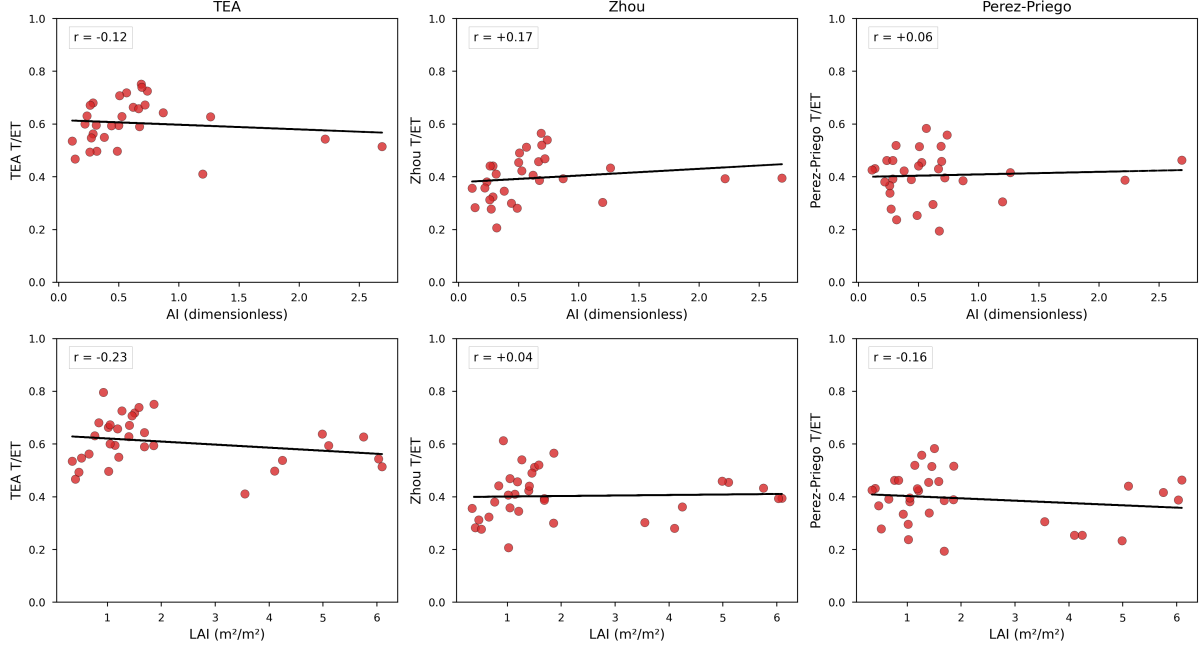


Figure 5: Scatter plots of site-mean T/ET against computed Aridity Index ($AI = P/PET$, dimensionless) and MODIS-derived Leaf Area Index (LAI , m²/m²). Lower AI values indicate drier conditions.

4 Discussion and Conclusions

By successfully replicating the analytical framework of Nelson et al. (2020), this analysis confirms that the discrepancies between data-driven partitioning methods are systematic and geographically consistent across Australian ecosystems.

The Random Forest model (TEA), trained exclusively on dry-canopy conditions, assumes evaporation is minimal; when forced to extrapolate to wet periods, it produces conservative evaporation estimates, yielding high residual T/ET. Conversely, physiological methods rely strictly on GPP and VPD constraints, predicting lower overall transpiration.

Environmental controls are evident: sites with higher LAI (> 3 m²/m²) show elevated T/ET regardless of method, consistent with increased canopy interception and reduced soil evaporation. Similarly, wetter sites ($AI > 0.65$) maintain higher T/ET, while arid sites ($AI < 0.2$) show the largest inter-method divergence, highlighting uncertainty in partitioning under water-limited conditions.

Despite these absolute magnitude differences, the methods exhibit powerful temporal synchronicity, proving they all reliably track physiological responses to water and carbon cycling across the Australian continent.

Data Availability

Site metadata, MODIS LAI climatology, and processed NetCDF archives are located at:

`/home/sanjays/et97_scratch2/oldscratch/Ozflux_data_full/OWUS_australia_agent_3_scientific/san,
data creation/`